

```
In [12]: import pandas as pd
```

```
In [15]: data = pd.read_csv("/home/student/Downloads/archive (3)/Iris.csv")
```

```
In [16]: data
```

Out[16]:

|  | Id  | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species |                |
|--|-----|---------------|--------------|---------------|--------------|---------|----------------|
|  | 0   | 1             | 5.1          | 3.5           | 1.4          | 0.2     | Iris-setosa    |
|  | 1   | 2             | 4.9          | 3.0           | 1.4          | 0.2     | Iris-setosa    |
|  | 2   | 3             | 4.7          | 3.2           | 1.3          | 0.2     | Iris-setosa    |
|  | 3   | 4             | 4.6          | 3.1           | 1.5          | 0.2     | Iris-setosa    |
|  | 4   | 5             | 5.0          | 3.6           | 1.4          | 0.2     | Iris-setosa    |
|  | ... | ...           | ...          | ...           | ...          | ...     | ...            |
|  | 145 | 146           | 6.7          | 3.0           | 5.2          | 2.3     | Iris-virginica |
|  | 146 | 147           | 6.3          | 2.5           | 5.0          | 1.9     | Iris-virginica |
|  | 147 | 148           | 6.5          | 3.0           | 5.2          | 2.0     | Iris-virginica |
|  | 148 | 149           | 6.2          | 3.4           | 5.4          | 2.3     | Iris-virginica |
|  | 149 | 150           | 5.9          | 3.0           | 5.1          | 1.8     | Iris-virginica |

150 rows × 6 columns

```
In [17]: data.describe()
```

Out[17]:

|       | Id         | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm |
|-------|------------|---------------|--------------|---------------|--------------|
| count | 150.000000 | 150.000000    | 150.000000   | 150.000000    | 150.000000   |
| mean  | 75.500000  | 5.843333      | 3.054000     | 3.758667      | 1.198667     |
| std   | 43.445368  | 0.828066      | 0.433594     | 1.764420      | 0.763161     |
| min   | 1.000000   | 4.300000      | 2.000000     | 1.000000      | 0.100000     |
| 25%   | 38.250000  | 5.100000      | 2.800000     | 1.600000      | 0.300000     |
| 50%   | 75.500000  | 5.800000      | 3.000000     | 4.350000      | 1.300000     |
| 75%   | 112.750000 | 6.400000      | 3.300000     | 5.100000      | 1.800000     |
| max   | 150.000000 | 7.900000      | 4.400000     | 6.900000      | 2.500000     |

```
In [18]: data.shape
```

Out[18]: (150, 6)

```
In [22]: data['Species'].value_counts()
```

Out[22]: Species  
Iris-setosa 50  
Iris-versicolor 50  
Iris-virginica 50  
Name: count, dtype: int64

```
In [26]: print(data.columns.tolist())
```

```
['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm', 'Species']
```

```
In [38]: data['SepalLengthCm'].mean()  
data['SepalWidthCm'].mean()  
data['PetalWidthCm'].mean()
```

```
Out[38]: 1.1986666666666668
```

```
In [39]: data['SepalLengthCm'].median()  
data['SepalWidthCm'].median()  
data['PetalWidthCm'].median()
```

```
Out[39]: 1.3
```

```
In [40]: data['SepalLengthCm'].std()  
data['SepalWidthCm'].std()  
data['PetalWidthCm'].std()
```

```
Out[40]: 0.7631607417008411
```

```
In [41]: data['SepalLengthCm'].var()  
data['SepalWidthCm'].var()  
data['PetalWidthCm'].var()
```

```
Out[41]: 0.582414317673378
```

```
In [35]: data.groupby('Species')['SepalLengthCm'].agg(['mean', 'max', 'median', 'std', 'var'])
```

```
Out[35]:
```

|                 | mean  | max | median | std      | var      |
|-----------------|-------|-----|--------|----------|----------|
| Species         |       |     |        |          |          |
| Iris-setosa     | 5.006 | 5.8 | 5.0    | 0.352490 | 0.124249 |
| Iris-versicolor | 5.936 | 7.0 | 5.9    | 0.516171 | 0.266433 |
| Iris-virginica  | 6.588 | 7.9 | 6.5    | 0.635880 | 0.404343 |

```
In [36]: data.groupby('Species')['PetalLengthCm'].agg(['mean', 'max', 'median', 'std', 'var'])
```

```
Out[36]:
```

|                 | mean  | max | median | std      | var      |
|-----------------|-------|-----|--------|----------|----------|
| Species         |       |     |        |          |          |
| Iris-setosa     | 1.464 | 1.9 | 1.50   | 0.173511 | 0.030106 |
| Iris-versicolor | 4.260 | 5.1 | 4.35   | 0.469911 | 0.220816 |
| Iris-virginica  | 5.552 | 6.9 | 5.55   | 0.551895 | 0.304588 |

```
In [37]: data.groupby('Species')['PetalWidthCm'].agg(['mean', 'max', 'median', 'std', 'var'])
```

```
Out[37]:
```

|                 | mean  | max | median | std      | var      |
|-----------------|-------|-----|--------|----------|----------|
| Species         |       |     |        |          |          |
| Iris-setosa     | 0.244 | 0.6 | 0.2    | 0.107210 | 0.011494 |
| Iris-versicolor | 1.326 | 1.8 | 1.3    | 0.197753 | 0.039106 |
| Iris-virginica  | 2.026 | 2.5 | 2.0    | 0.274650 | 0.075433 |

```
In [42]: data.groupby('Species')['SepalWidthCm'].agg(['mean', 'max', 'median', 'std', 'var'])
```

Out[42]:

|                 | mean  | max | median | std      | var      |
|-----------------|-------|-----|--------|----------|----------|
| Species         |       |     |        |          |          |
| Iris-setosa     | 3.418 | 4.4 | 3.4    | 0.381024 | 0.145180 |
| Iris-versicolor | 2.770 | 3.4 | 2.8    | 0.313798 | 0.098469 |
| Iris-virginica  | 2.974 | 3.8 | 3.0    | 0.322497 | 0.104004 |

In [43]:

In [ ]: